

Zhaorui Meng

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Profile

I am a Ph.D. student at Xiamen University, advised by Shihui Guo, with a mathematical background in algebra, graph theory and numerical methods. My research focuses on human motion modeling: sparse inertial capture, physics-based imitation and motion quality evaluation. I am also interested in vision-language-action (VLA) and world action models (WAM), particularly how human motion priors and physical structure can support action learning and planning.

Education

Xiamen University Ph.D., School of Informatics Advisor: Shihui Guo Expected graduation: 2028	2025 - Present
Xiamen University Master's studies in Pure Mathematics Algebraic graph theory; transitioned to the Ph.D. program in 2025.	2023 - 2025
Xiamen University B.Sc. Information and Computational Mathematics	2019 - 2023

Research

Improving Sparse IMU-based Motion Capture with Motion Label Smoothing

Zhaorui Meng, Lu Yin, Yangqing Hou, Anjun Chen, Shihui Guo, Yipeng Qin

AAAI 2026 | [Paper](#) | [Code](#)

Sparse inertial observations leave full-body motion ambiguous, but label regularization must respect continuous rotations and spatiotemporal correlations: independent noise can break the structure a capture model needs to learn. This work uses skeleton-aware Perlin perturbations in a continuous 6D rotation representation, coupling nearby frames and joints along skeletal chains.

Fourier and power-spectrum analysis characterize the low-frequency perturbations, treating physically structured variation as a training prior rather than adding architectural complexity. Cross-model evaluation examines whether this prior transfers across capture architectures. The central idea is to regularize motion labels through their geometry and correlations, rather than treating each joint and frame as an independent target.

Distinguishing Imitation Error from Intrinsic Motion Learning Difficulty

Zhaorui Meng, Lu Yin, Xinrui Chen, Chengxu Zuo, Anjun Chen, Shihui Guo, Yipeng Qin

ICML 2026 | [Paper](#) | [Code](#)

Imitation error confounds a policy's ability with the intrinsic sensitivity of the reference motion: the same tracking deviation can reflect different control demands. Raw movement speed or torque magnitude is not equivalent to difficulty; small pose variations can require large torque adjustments. This motivates examining the local torque landscape through rigid-body dynamics Jacobians. This reframes difficulty in terms of the torque response to motion perturbations, rather than the visible size of the motion itself.

Torque Variation Score (TVS) combines complementary aspects of this landscape: SVD-based spectral diversity captures variation across sensitivity directions, joint-wise variance summarizes how sensitivity varies, and segment-level aggregation accounts for temporal structure. Defined independently of an imitation policy, TVS supports difficulty-stratified evaluation rather than using a policy's own error to label its test motions. MID and DSJE reveal capability differences that aggregate tracking error can hide, distinguishing the difficulty a policy can imitate from its errors at different difficulty levels. Curriculum experiments then connect evaluation to learning by using motion difficulty to organize training, exploring whether this diagnosis can also guide how policies acquire motion skills.

Research (continued)

CogEval: A Cognition-aware Evaluator for Human Motion Quality Assessment

Zhaorui Meng, Yile Pan, Lu Yin, Chengxu Zuo, Anjun Chen, Shihui Guo, Yipeng Qin

Under review | [Overview](#)

Familiarity does not establish physical plausibility, while physical checks alone do not establish perceptual naturalness. CogEval combines complementary, reference-free judgments: KNN memory retrieval provides an experience-based assessment, and analytical checks examine stability, contact and body structure. Joint-velocity distribution entropy characterizes motion complexity; regularized fusion balances the two sources of evidence. This dual-system design evaluates generated motion without a paired ground-truth trajectory, making room for both learned preferences and explicit physical reasoning.

Automated Task Scheduling for Cloth and Deformable Body Simulations in Heterogeneous Computing Environments

Chengzhu He, Zhendong Wang, **Zhaorui Meng**, Junfeng Yao, Shihui Guo, Huamin Wang

SIGGRAPH 2025 | [Paper](#) | [Code](#)

In this collaborative simulation framework, GPU compute gains on small tasks can be outweighed by transfer overhead, making task granularity and dependencies central to heterogeneous scheduling. My main contribution was task scheduling: representing dependencies as a directed acyclic graph and working on Heterogeneous Earliest Finish Time (HEFT) scheduling and task merging to balance computation against communication costs. This design considers where tasks execute together with how much work each task contains. I also provided minor assistance with simulation-engine setup.

Towards Physical Intent-Driven General Motion Imitation in Humanoid Teleoperation

Lu Yin, Xinrui Chen, **Zhaorui Meng**, Haodong Zhang, Lili Liu, Qi Ye, Shihui Guo, Anjun Chen

Under review | [Paper](#) | [Project](#)

The framework models physical intent rather than simply tracking potentially flawed human motion references. Assisted with parts of the teleoperation pipeline, connecting five-IMU motion estimation, retargeting and control execution.

Research Experience

Cardiff University | Remote research

Oct 2025 - May 2026

Research under the supervision of Yipeng Qin.

Awards

Xinghao Scholarship

March 2026

Bailu Scholarship

May 2023

Academic Scholarships

Sept 2022 / Sept 2020

Technical Skills

Python, PyTorch, NumPy and SciPy; working knowledge of C++.

Languages

English: IELTS 6.5.